Type MS · Explain how inhibitors or a mutation change enzyme activity

Answer 1 of 5 · Question 3 part b · 2 marks

2023 JUN · Paper 01 · extract 1 of 2 · find part b in this mark scheme extract

Question number					
3	(a)	(i)	• time in minutes; • 0-10, 11-20, 21-30, 31-40,41-50,51-60;		2
		(ii)	• 75 + 72 + 68; plus/minus 2 for each one • 215 cm^3;	ECF	2
		(iii)	• rapid reaction/a lot/large amount of gas produced, at start; • rate slows/less gas (for next 10 minutes); • drops a lot/little gas produced (after 40 minutes);		3
		(iv)	any three from • at start plenty of substrate/many E-S complexes formed; • substrate used up/less substrate available; • reaction slows/less gas/product, produced/less enzyme action; • most substrate used at end so little gas produced;		3
	(b)		• less gas produced/reaction lower/less; • because active sites blocked;		2

Total 12

Type MS · Explain how inhibitors or a mutation change enzyme activity

Answer 1 of 5 · Question 3 part b · 2 marks

2023 JUN · Paper 01 · extract 2 of 2 · find part b in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain how inhibitors or a mutation change enzyme activity

Answer 2 of 5 · Question 5 part b.iii · 2 marks

2023 NOV · Paper 01 · extract 1 of 2 · find part b.iii in this mark scheme extract

Question number				
5	(a)	4^3; = 64;	Full marks for correct answer	2
	(b) (i)	• change in order/sequence; • of bases; • different protein may be formed;		3
	(ii)	uv light/λ rays/ionising radiation/X-rays;	accept any suitable example	1
	(iii)	• can't catalyse reaction/slows reaction/activity; • because can't form E-S complexes/substrate can't bind with active site/active site not complementary;		2
	(iv)	doesn't affect working of enzyme/mutation does not change amino acid/doesn't affect structure/shape of enzyme/different coding for same amino acid;		1

Total 9 marks

Type MS · Explain how inhibitors or a mutation change enzyme activity

Answer 2 of 5 · Question 5 part b.iii · 2 marks

2023 NOV · Paper 01 · extract 2 of 2 · find part b.iii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain how inhibitors or a mutation change enzyme activity

Answer 3 of 5 · Question 9 part c · 4 marks

2019 JUN · Paper 02 · extract 1 of 2 · find part c in this mark scheme extract

9 (a) (i)	• reference to platelets; • fibrinogen converted to fibrin; • (fibrin) mesh/crosslinks/fibres formed;		3
(ii)	• reduced blood flow to heart/cardiac muscle / tissue/cells; • less oxygen/glucose delivered; • less (aerobic) respiration/energy released (by cardiac tissue);		3
(b)(i)	size of blood clot/volume of water /aspirin solution used/temperature/shape of blood clot/width of straw;	Allow valid alternatives	1
(ii)	use water without aspirin dissolved;		1
(iii)	• the more tablets dissolved the more solution collected; • the more tablets dissolved the quicker the solution is collected; • aspirin dissolves/acts on blood clot; • size of blood clot reduced more quickly with more tablets/blood clot dissolves/breaks down faster with more tablets;		Max 3
(c)	Any four from: • aspirin binds to enzyme/aspirin competes (with substrate) for active site; • active site of enzyme changed/blocked; • less/no substrate binds (to enzyme); • reaction involving release of chemicals reduced/stopped; • platelets do not stick together;		Max 4

Total question 9 = 15 marks

Type MS · Explain how inhibitors or a mutation change enzyme activity

Answer 3 of 5 · Question 9 part c · 4 marks

2019 JUN · Paper 02 · extract 2 of 2 · find part c in this mark scheme extract

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Type MS · Explain how inhibitors or a mutation change enzyme activity

Answer 4 of 5 · Question 2 part b.iii · 2 marks

2022 JUN · Paper 02 (regional R) · extract 1 of 2 · find part b.iii in this mark scheme extract

2 (a)	(i)	• a tube containing inhibitor and substrate; • but no enzyme;		2
	(ii)	any three from • concentration of enzyme; • concentration of inhibitor; • temperature; • pH; • same filter in colorimeter;		3
(b)	(i)	axes labelled with units; axes correct way round; plots;		3
	(ii)	suitable curve;		1
	(iii)	similar shaped curve; reaching maximum sooner;		2
	(c)	same maximum never reached; despite increase in substrate concentration;		2

Total 13 marks

Type MS · Explain how inhibitors or a mutation change enzyme activity

Answer 4 of 5 · Question 2 part b.iii · 2 marks

2022 JUN · Paper 02 (regional R) · extract 2 of 2 · find part b.iii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain how inhibitors or a mutation change enzyme activity

Answer 5 of 5 · Question 2 part c · 2 marks

2022 JUN · Paper 02 (regional R) · extract 1 of 2 · find part c in this mark scheme extract

2 (a)	(i)	• a tube containing inhibitor and substrate; • but no enzyme;		2
	(ii)	any three from • concentration of enzyme; • concentration of inhibitor; • temperature; • pH; • same filter in colorimeter;		3
(b)	(i)	axes labelled with units; axes correct way round; plots;		3
	(ii)	suitable curve;		1
	(iii)	similar shaped curve; reaching maximum sooner;		2
	(c)	same maximum never reached; despite increase in substrate concentration;		2

Total 13 marks

Type MS · Explain how inhibitors or a mutation change enzyme activity

Answer 5 of 5 · Question 2 part c · 2 marks

2022 JUN · Paper 02 (regional R) · extract 2 of 2 · find part c in this mark scheme extract

Question number	Answer	Notes	Marks